Simple Regression Analysis

$$y = \beta_0 + \beta_1 x + u$$

Estimation

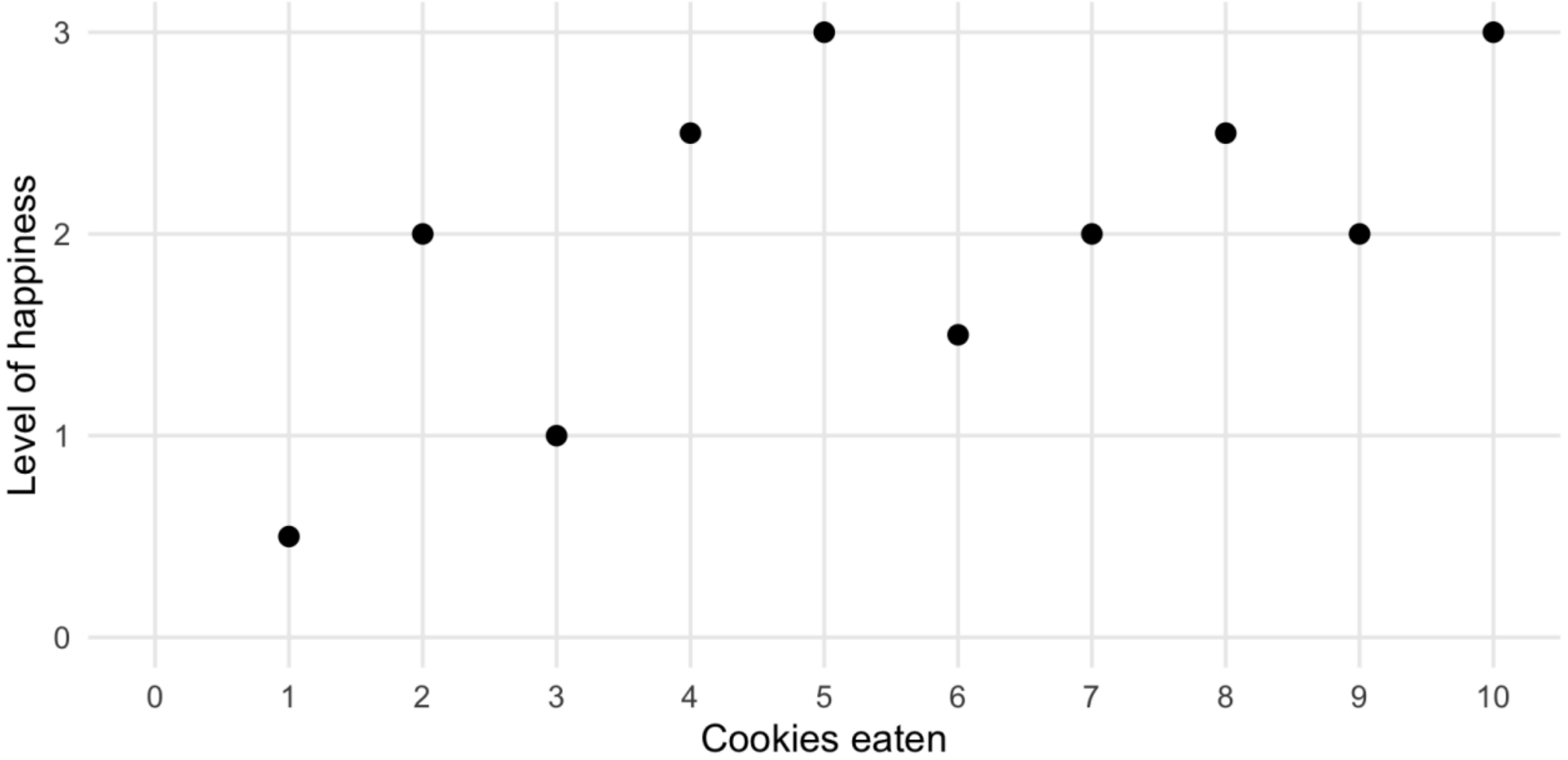
Population Model

We have a set of **data** that we want to model:

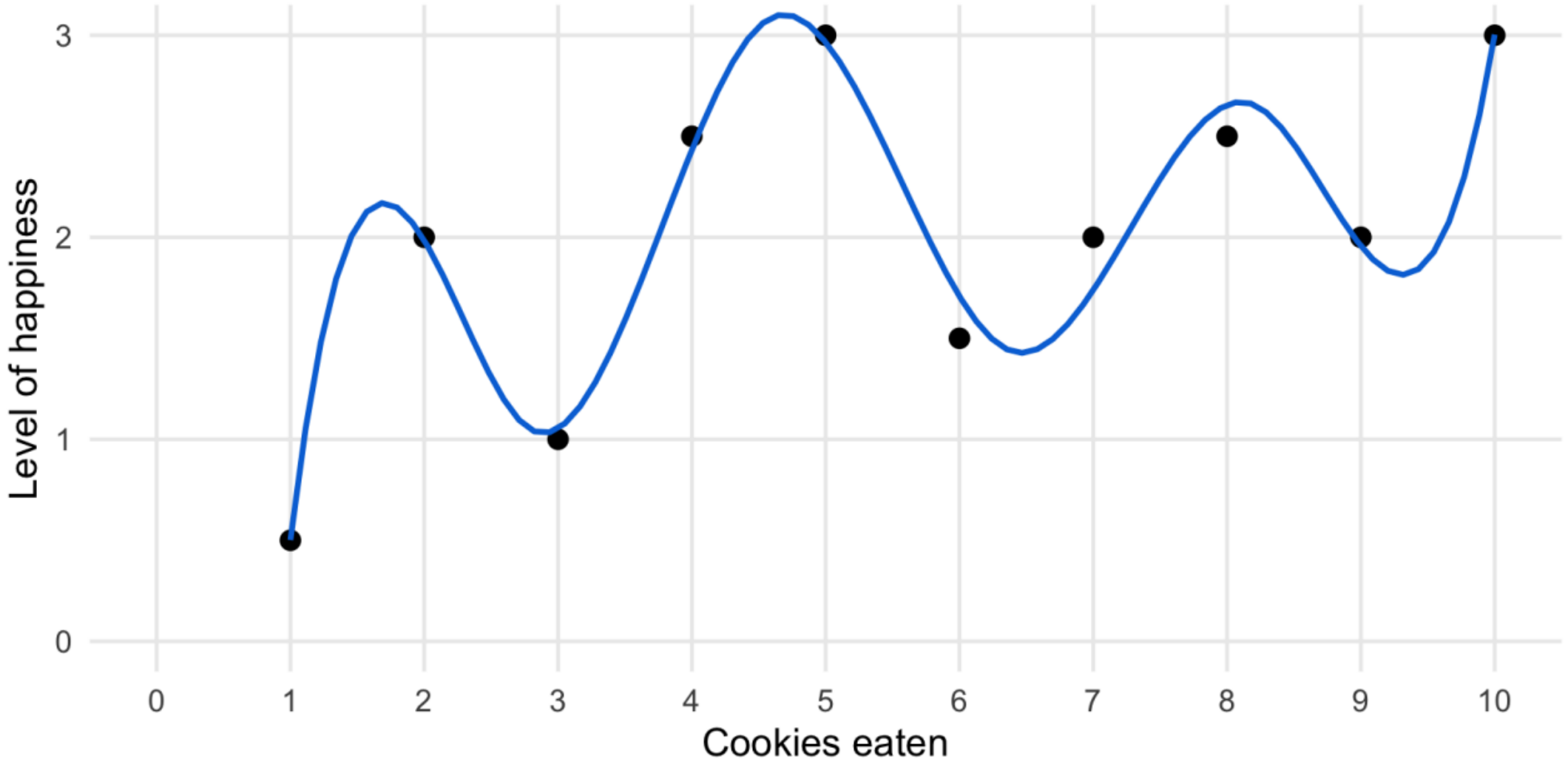
- y is the dependent variable (outcome)
- x is the independent variable
- We don't believe the relationship is **deterministic**, so we assume there is some randomness u.

- What's the simplest mathematical relationship between these two variables that could exist?

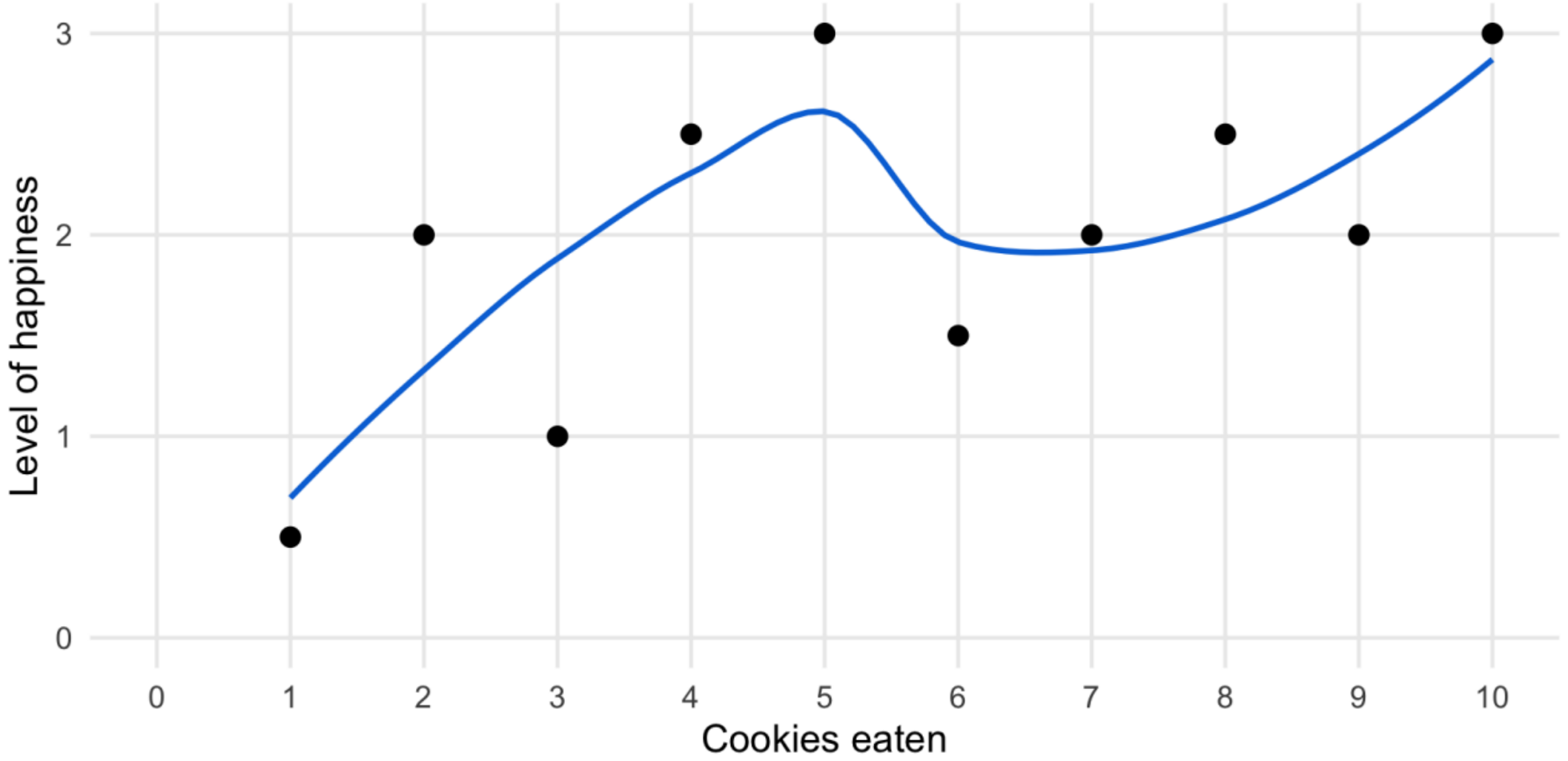
Population Model



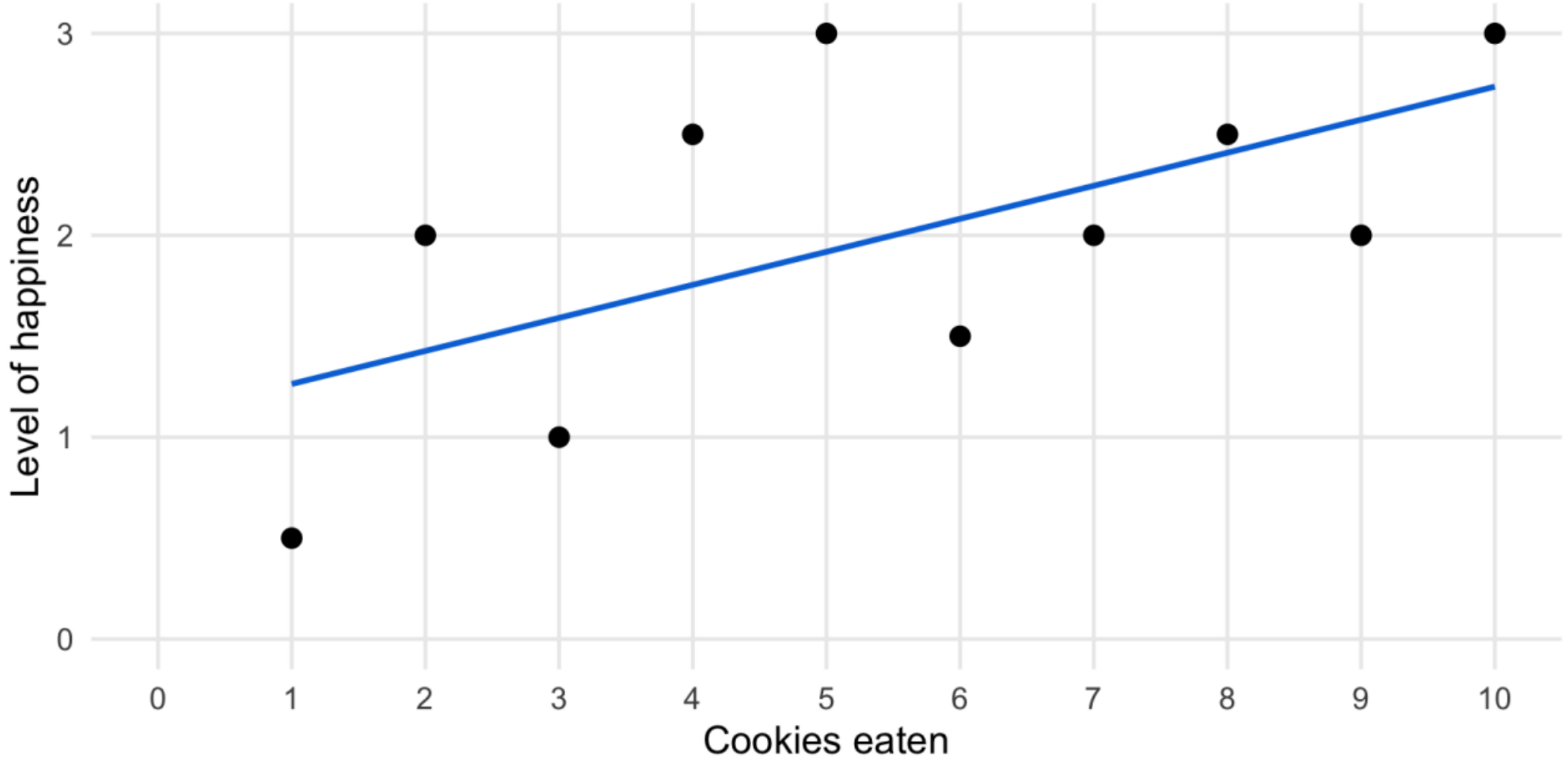
Population Model



Population Model



Population Model



Linear Regression Model

- This lecture introduces the linear regression model relating one variable, *X*, to another, *Y,* which postulates a *linear* relationship between *X* and *Y*.
- The simple regression model can be used to study the relationship between two variables.
- Learning how to interpret the simple regression model is good practice for studying multiple regression, which we will do in subsequent lectures.

Simple Regression Model

- For the simple regression model, it has only one regressor x:

$$y = E(y|x) + u = \beta_0 + \beta_1 x + u$$

- y is the dependent variable (outcome)
- x is the regressor, or independent variable
- u is an error term to allow for random variation in y not captured by x. You can usefully think of u as standing for "unobserved".

Simple Regression Model

- For the simple regression model, it has only one regressor x:

$$y = E(y|x) + u = \beta_0 + \beta_1 x + u$$

- y is the dependent variable (outcome)
- x is the regressor, or independent variable
- u is an error term to allow for random variation in y not captured by x.
- β's are called **coefficients**:
  - β_0: intercept parameter, sometimes called the constant term
  - β_1: slope parameter

Some Terminology

- In (linear) regression models, we typically call

y	x
Dependent Variable	Independent Variable
Left-Hand Side Variable	Right-Hand Side Variable
Explained Variable	Explanatory Variable
Regressand	Regressor
	Covariate, Control Variables

Interpretation of the Coefficients

- For the simple regression model, it has only one regressor x:

$$y = E(y|x) + u = \beta_0 + \beta_1 x + u$$

- This model addresses the issue of the functional relationship between y and x. If the other factors in u are held fixed, so that the change in u is zero, then x has a *linear* effect on y:

$$\Delta y = \beta_1 \Delta x$$

Interpretation of the Coefficients

- However, the most difficult issue to address is whether the model really allows us to draw **ceteris paribus** conclusions about how x affects y.
- We just saw that β_1 *does* measure the effect of x on y, holding all other factors (in u) fixed. Is this the end of the **causality issue**? Unfortunately, no.
- The point is how unobserved u is related to x.

Population Model: Assumptions

Zero Conditional Mean Assumption:

$$\mathrm{E}[u|x] = \mathrm{E}[u] = 0.$$

- For the latter one, i.e. $\mathrm{E}[\mu] = 0$ says nothing about the relationship between u and x, but simply makes a statement about the distribution of the unobserved factors in the population.
- Without loss of generality, we can assume that things such as average ability are zero in the population of all working people.

Population Model: Assumptions

Zero Conditional Mean Assumption:

$$\mathrm{E}[u|x] = \mathrm{E}[u] = 0.$$

- But $\mathrm{E}[\mu|x] = 0$, this is often a **stringent assumption.**
- Consequence: $\mathrm{E}[y|x] = \beta_0 + \beta_1 x \rightarrow$ causal effect !
- β_0 and β_1 are interesting population parameters and have their interpretations, but UNKNOWN.

Population Model: Assumptions

- $E(u|x) = 0$ implies that the disturbance u contains no systematic information of x that can be used to predict the mean of y.
- Thus, the conditional mean assumption implies that x and u are uncorrelated (but not vice versa, why?). Or, if x and u are correlated, then the assumption is violated.
- Without this assumption, we can never infer the causal effect of x on y (what's the main goal of econometrics?) !

Population Model: Assumptions

- Of course, the possibility that x is correlated with u is almost always a concern in simple regression analysis with **observation** or **nonexperimental** data.
- Using simple regression when u contains factors affecting y that are also correlated with x can result in *spurious correlation*: that is, we find a relationship between y and x that is really due to other unobserved factors that affect y and also happen to be correlated with x. We will further discuss this issue in multiple regression.

Estimating a Regression: Intuition

- Given the following two assumptions:

✓ Mean zero error: $\mathbb{E}[u] = 0$. This is *without loss of generality, given intercept.*

✓ Mean independence: $\mathbb{E}[u|x] = \mathbb{E}[u] = 0$ for all values of x.

  - This is often a **critical assumption.**

Then we can use a system of equations to solve $\hat{\beta}$:

1. Mean zero error: $\mathbb{E}[y - \beta_0 - \beta_1 x] = 0$
2. Mean independence: $\mathbb{E}[u|x] = x(y - \beta_0 - \beta_1 x) = 0$

Estimating a Regression

- Given a sample of data, we have:

$$n^{-1}\sum_{i=1}^{n}\left(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i\right) = 0$$

$$n^{-1}\sum_{i=1}^{n} x_i\left(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i\right) = 0$$

Estimating a Regression

- Using the basic properties of the summation operator:

$$\bar{y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x}$$

  - where $\bar{y} = n^{-1} \sum_{i=1}^{n} y_i$ is the sample average of the y_i and likewise for $\bar{x}$.
  - Therefore, once we have the slope estimate $\hat{\beta}_1$, it is straightforward to obtain the intercept estimate $\hat{\beta}_0$.

Estimating a Regression

- We can solve this system for the desired values of β_1:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n} x_i(y_i - \bar{y})}{\sum_{i=1}^{n} x_i(x_i - \bar{x})} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

Given the summation's properties:

$$\sum_{i=1}^{n} x_i(y_i - \bar{y}) = \sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})$$

$$\sum_{i=1}^{n} x_i(x_i - \bar{x}) = \sum_{i=1}^{n}(x_i - \bar{x})^2$$

Estimating a Regression

- The estimates given above are called the **ordinary least squares (OLS)** estimates of β_0 and β_1.
- To justify this name, for any $\hat{\beta}_0$ and $\hat{\beta}_1$ define a fitted value:

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

- There is a fitted value for each observation in the sample. The **residual** for observation i is the difference between the actual y_i and its fitted value.

Ordinary Least Squares (OLS)

Definition: The residual sum of squares (SSR):

$$\mathbf{SSR} = \sum_{i=1}^{N} \hat{u}_i^2 = \sum_{i=1}^{N} (y_i - \hat{y}_i)^2$$

with $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$.

- The name OLS comes from the fact that it returns β_0, β_1 that minimize SSR given all observed data!
- Sidenote: $\hat{u}$ is called a *residual* (importantly different from an error term)

Ordinary Least Squares (OLS)

- OLS (minimizing SSR):

$$\min_{\{b_0,b_1\}} \sum_{i=1}^{n} (Y_i - b_0 - b_1 X_i)^2$$

We can readily obtain its solution with first-order conditions, as

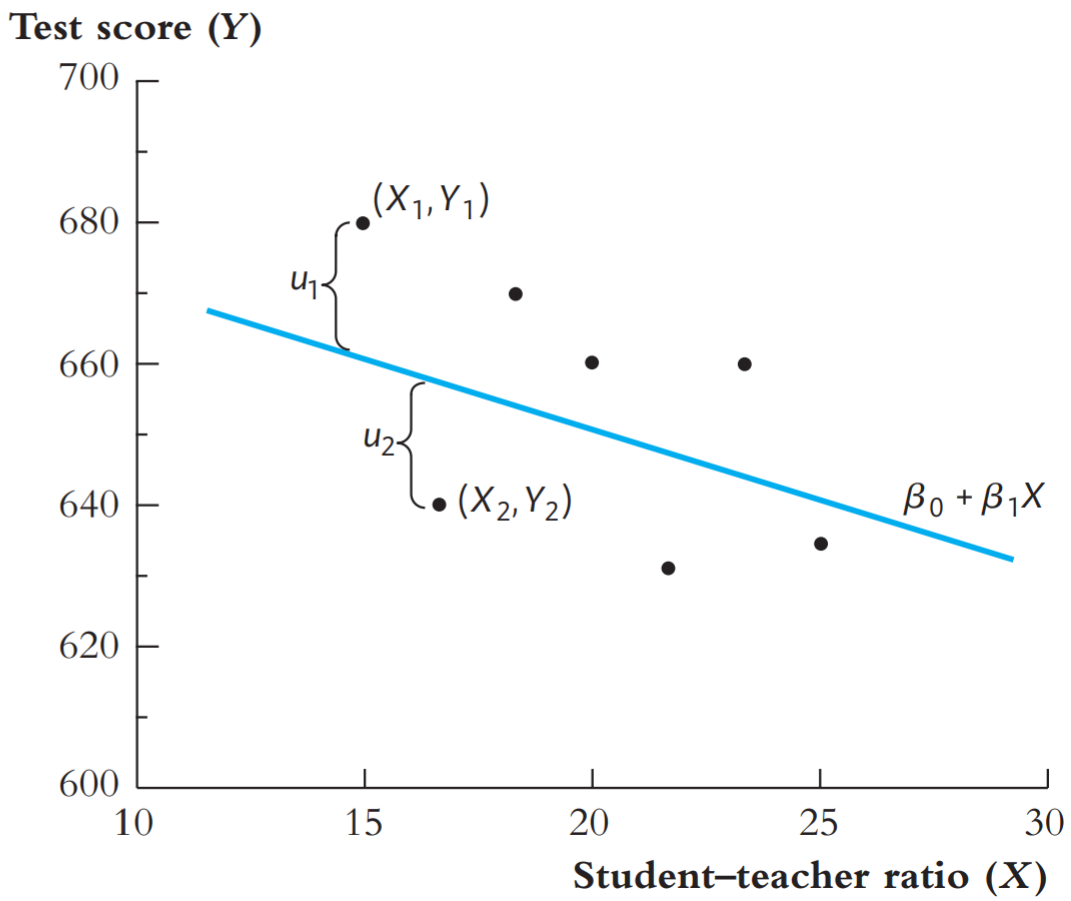
$$\hat{\beta}_{2\times 1} = \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{pmatrix} = \begin{pmatrix} \bar{Y} - \hat{\beta}_1 \bar{X} \\ \sum_{i=1}^{n} (X_i - \bar{X}) Y_i / \sum_{i=1}^{n} (X_i - \bar{X})^2 \end{pmatrix},$$

where

$$\bar{X} = n^{-1} \sum_{i=1}^{n} X_i \text{ and } \bar{Y} = n^{-1} \sum_{i=1}^{n} Y_i.$$

Exp. Ordinary Least Squares (OLS)

The scatterplot shows hypothetical observations for seven school districts. The population regression line is $\beta_0 + \beta_1 X$. The vertical distance from the ith point to the population regression line is $Y_i - (\beta_0 + \beta_1 X_i)$, which is the population error term u_i for the ith observation.



Goodness of Fit

- There are **two separate things** we care about when we run a regression:
  1. Is the model fitting the data?
  2. Is the model capturing a causal parameter we care about?

Goodness of Fit

To check on ①, we use **goodness of fit measures**. We can think of each observation as being made up of an explained part, and an unexplained part $y_i = \hat{y}_i + \hat{u}_i$:

- SST (Sum of Squares Total)

$$SST = \sum_{i=1}^{n} (y_i - \bar{y})^2$$

Goodness of Fit

To check on ①, we use **goodness of fit measures**. We can think of each observation as being made up of an explained part, and an unexplained part $y_i = \hat{y}_i + \hat{u}_i$:

- SSE (Sum of Squares Explained)

$$SSE = \sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2$$

Goodness of Fit

To check on ①, we use **goodness of fit measures**. We can think of each observation as being made up of an explained part, and an unexplained part $y_i = \hat{y}_i + \hat{u}_i$:

- SSR (Sum of Squared Residuals)

$$SSR = \sum_{i=1}^{n} \hat{u}_i^{\,2}$$

Goodness of Fit

To check on ①, we use **goodness of fit measures:**

- Then, we define the **fraction of SST explained by our regression,** call this the R-squared of regression**:**

$$R^2 = \frac{SSE}{SST} = 1 - \frac{SSR}{SST}$$

Goodness-of-Fit (continued)

- We can also think of R^2 as being equal to the squared correlation coefficient between the actual y_i and the values $\hat{y}_i$:

$$R^2 = \frac{\left(\sum (y_i - \bar{y})\left(\hat{y}_i - \bar{\hat{y}}\right)\right)^2}{\left(\sum (y_i - \bar{y})^2\right)\left(\sum \left(\hat{y}_i - \bar{\hat{y}}\right)^2\right)} = \rho^2$$

Some notes on R^2

- Always ranges between 0 and 1 (0 = no relationship; 1 = perfect line)
- Related to the correlation between $\hat{y}$ and y – hence, *not* indicative of causality!
- Even low R^2 models can have strong causal interpretations – be clear on what your research goal is!
  - Again, R^2 says nothing about $E(u|x) = 0$!

Unbiasedness: Estimating Causal POI

- Suppose that we have a correctly identified model, such that β_1 is a true causal parameter of interest (POI)

How do we know the regression gives us β_1?

- $\hat{\beta}$ itself is random, and its realization depends on the dataset you collect.
- Unbiasedness: we want the distribution of $\hat{\beta}$ to be centered at β.
- What do we need to assume in order for $\mathrm{E}[\hat{\beta}] = \beta$?

Unbiasedness: Estimating Causal POI

Assumptions:

A1. Population model is linear in parameters

$$\mathbf{y} = \beta_0 + \beta_1 x + u.$$

A2. A random sample of size n. That is, $\{(x_i, y_i): i=1, 2, \ldots, n\}$ are independently and identically distributed (i.i.d.), each drawn from the population model, satisfying $y_i = \beta_0 + \beta_1 x_i + u_i$.

Unbiasedness: Estimating Causal POI

A3. Zero conditional mean condition: $\boldsymbol{E(u|x) = 0}$, implying that the explanatory variable x is conditional exogenous.

A4. Sample variation: x is not a constant, and has variation across observations.

Under the above 4 assumptions, we have $\mathbf{E}[\widehat{\boldsymbol{\beta}}] = \boldsymbol{\beta}$.

Unbiasedness: Estimating Causal POI

- Remember that unbiasedness is a feature of the sampling distributions of $\hat{\beta}_0$ and $\hat{\beta}_1$, which says nothing about the estimate that we obtain for a given sample.
- We hope that, if the sample we obtain is somehow "typical", then our estimate should be "near" the population value.
- Unfortunately, it is always possible that we could obtain an unlucky sample that would give us a point estimate far from $\hat{\beta}_1$.

Variance of the OLS Estimators

- In addition to knowing that the sampling distribution of $\hat{\beta}$ is centered about β ($\hat{\beta}$ is thus unbiased), it is important to know **how much dispersion the distribution of** $\hat{\beta}$ has around its center β.
- The smaller the dispersion, the better?

Variance of the OLS Estimators

- The smaller the dispersion, the better?
- How to compute the variance or standard deviations of $\hat{\beta}$, namely, $Var(\hat{\beta})$ or $sd(\hat{\beta})$?
  - According to the definition of variance, the square root of $Var(\hat{\beta})$ is equal to $sd(\hat{\beta})$

Variance of the OLS Estimators

- It turns out that the variance of the OLS estimators can be computed under an addition assumption, which states that the variance of the error term, μ, conditional on x, is constant. This is known as the **conditional homoskedasticity** assumption.

$$Var[u|x] = \sigma^2$$

  - σ^2 is often called the error variance. It is thus a measure of the spread of the observations around the regression line.

Variance of the OLS Estimators

- Then we can show that, the **standard deviation of $\widehat{\boldsymbol{\beta}}_1$**, is:

$$sd(\widehat{\boldsymbol{\beta}}_1) = \frac{\sigma}{\sqrt{SST_x}}$$

  - where $SST_x = \sum_{i=1}^{n}(x_i - \bar{x})^2$.
- Important to note that $sd(\hat{\beta})$ can be not computed directly because it depends on σ, the unknown parameter!

Variance of the OLS Estimators

- The estimate for the error variance σ^2 is:

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^{n} \hat{u}_i^2}{n-2} = \frac{SSR}{n-2}$$

- The reason for using the divisor $n-2$ here is to correct for a slight bias introduced because two regression coefficients were estimated.
- It can be shown that

$$E(\hat{\sigma}^2) = \sigma^2$$

that is, $\hat{\sigma}^2$ is an unbiased estimator of σ^2.

Variance of the OLS Estimators

- For now, our primary interest is in using $\hat{\sigma}^2$ to estimate the standard deviations of $\hat{\beta}$, which is also called the **standard error of $\hat{\beta}$**:

$$se(\hat{\beta}) = \frac{\hat{\sigma}}{\sqrt{SST_x}}$$

This will be useful for hypothesis testing, which will be seen in more details in the later lecture.

The Gauss-Markov Theorem

- We know one justification for OLS already: under Assumption A1-A4, OLS is unbiased.
- Recall the four assumptions for unbiasedness:
  - **A1: Population model is *linear* in parameters**
  - **A2: We can use a *random sample* from the population model**
  - **A3: Mean independence $\mathbf{E}[u|x] = 0$, implying that all of the explanatory variables are *conditional exogenous***
  - **A4: Sample variation:** x has variation across observations.

The Gauss-Markov Theorem

- In fact, there could be *many* unbiased estimators of β under these four assumptions. Might there be other unbiased estimators with variances smaller than the OLS estimators (Good thing)?
- The four assumptions for unbiasedness (A1-A4), plus the **homoskedasticity assumption** are known as the *Gauss-Markov* assumptions. Under these five assumptions, the OLS estimator $\hat{\beta}$ has the smallest variance, among all of those unbiased linear estimators.

The Gauss-Markov Theorem

- In sum, given the Gauss-Markov assumptions, it can be shown that OLS is "BLUE"
  - Best
  - Linear
  - Unbiased
  - Estimator
- Thus, if these assumptions hold, use OLS, because it provides an **unbiased** and **efficient** linear estimates.

Conclusion

- We demonstrated the algebra of the OLS regression line, including slope and intercept parameters in the population model, computation of fitted values, residuals and formulas for the sampling variances of the OLS estimators.
- We showed that, under some assumptions, the OLS estimators are unbiased.
- The key assumption for causal inference is that the error term has zero mean, or average, given any value of the independent variable.

- Next time: Linear regression with multiple regressors.